# D1 Control Tower: the code, each file followed by its flowchart

The eight exhibits E1 to E8 exactly as they are in **d1/** of the kaveri-dock-lab repository. The code pages are generated from the files themselves, so what you read here is what runs. Plain, step-numbered Python with comments; **claude-fable-5-1**, the latest model, for every call, with the API's server-side fallback switched on. After each file comes a colour flowchart of the same code, with the points that matter for the CCAR-P exam marked by red dashed callouts.

## How the files fit together

**common.py** is the shared desk: the two switches, the model name, the three inputs read from data/, the two tool schemas, the two tool functions, and the helpers every exhibit calls (text_of, trace, make_client, ask, ask_json, ask_async). It is chapter 0 of this document. **standin.py** is the scripted stand-in that answers when DRY_RUN=1; it is printed in the appendix and you do not need it to learn the patterns. E3 reuses E2's classify and gate. E8 reuses E5's make_plan, run_worker and synthesise.

## Running the files

```
# live, from the repo root, with ANTHROPIC_API_KEY in a .env file there (git-ignored):
pip install anthropic
cd d1
python e1_augmented_llm.py

# on battery (no key, no cost), Windows Command Prompt:
set DRY_RUN=1
python e1_augmented_llm.py
set DRY_RUN=

# on battery, Linux or Mac:
DRY_RUN=1 python e1_augmented_llm.py

# everything at once on battery:   run_all.bat  (Windows)   or   ./run_all.sh
# E7 branches on battery:   set D1_SCENARIO=truncate   (or refuse, exhaust) before running e7
```

## How to read the flowcharts

| Colour | Means |
|---|---|
| Yellow box | Code setting things up: constants, schemas, the tool functions. The model never sees these. |
| Green box | A model call. Each one is paid for live. Count the green boxes to count the bill. |
| Blue box | A code step: building the conversation, running a tool, counting, a plain-Python gate. |
| Red diamond | A decision on one field, usually stop_reason or a word against a set. |
| Purple box, light | A result. |
| Purple box, dark border | A hand-off to a human, or a stop. |
| Red dashed callout with a star | An exam point: the fact, law, or distractor a stem is likely to test. |

## Status

All eight files, and E7 under its three scenarios, run on battery with the traces shown in d1/README.md. E1 was run live against the API from your laptop with the earlier version of this file and produced the same two-turn trace. The rewritten files have not yet been run live; the control flow is unchanged, the wording of a live reply will vary.

## Contents

0. common.py, the shared desk (code only)

1. E1 The augmented LLM (`e1_augmented_llm.py`) code, then flowchart

2. E2 Prompt chaining (`e2_chaining.py`) code, then flowchart

3. E3 Routing (`e3_routing.py`) code, then flowchart

4. E4 Parallelization (`e4_parallel.py`) code, then flowchart

5. E5 Orchestrator-workers (`e5_orchestrator_workers.py`) code, then flowchart

6. E6 Evaluator-optimizer (`e6_evaluator_optimizer.py`) code, then flowchart

7. E7 The agent loop (`e7_agent_loop.py`) code, then flowchart

8. E8 Re-planning (`e8_replanning.py`) code, then flowchart

Appendix. standin.py, the dry-run stand-in (code only)

# Chapter 0 common.py: the shared desk

Read STEP A to STEP F once. Every exhibit that follows imports from here, so when a chapter says 'ask comes from common.py' or 'the tool is fake_status', this is where it lives.

```
# common.py  -  the shared desk for the eight Domain 1 exhibits.
#
# Every exhibit imports from here, so the things that never change live in one place:
#   STEP A  the two switches (DRY_RUN, D1_SCENARIO) and the optional .env loader
#   STEP B  which models to use, and how long a reply may be
#   STEP C  the three inputs: Padma's email, the refund claim, Meera's brief (read from data/)
#   STEP D  the two tool DESCRIPTIONS (schemas) the model may ask for
#   STEP E  the tools THEMSELVES: plain Python that answers from a canned world
#   STEP F  helpers: text_of(), trace(), make_client(), ask(), ask_json(), ask_async()
#
# DRY_RUN=1  -> make_client() returns the scripted stand-in from standin.py:
#               same control flow, canned replies, no network, no bill.
# unset      -> the real anthropic SDK. Needs:  pip install anthropic
#               and ANTHROPIC_API_KEY in the environment, or in a .env file in the repo root (git-ignored).
import asyncio
import json
import os
import re
from pathlib import Path


# ---------------------------------------------------------------
# STEP A. Switches, and the optional .env loader
# ---------------------------------------------------------------

# If a .env file exists (next to this file, or in the repo root), copy its KEY=value lines into the
# environment. A variable that is already set wins. The file is git-ignored: never commit the key.
for env_file in (Path(__file__).with_name(".env"), Path(__file__).resolve().parent.parent / ".env"):
    if env_file.exists():
        for line in env_file.read_text().splitlines():
            line = line.strip()
            if line == "" or line.startswith("#") or "=" not in line:
                continue
            key, value = line.split("=", 1)
            os.environ.setdefault(key.strip(), value.strip().strip('"').strip("'"))
DRY_RUN = os.environ.get("DRY_RUN") == "1"              # battery mode: stand-in client, zero paid calls
SCENARIO = os.environ.get("D1_SCENARIO", "default")      # steers the stand-in's final reply in E7


# ---------------------------------------------------------------
# STEP B. Models and reply length
# ---------------------------------------------------------------

MODEL = "claude-fable-5-1"          # one model for everything: judgement, planning, drafting, one-word sorts
MAX_TOKENS = 128000                 # the longest reply we accept: the model's hard ceiling (128K output tokens)
# The API rejects anything above 128,000 with a 400. Thinking tokens count toward this cap, so a high
# cap means a cut-off reply is next to impossible. A short answer still costs only what it uses.
# Fable 5.1 is the latest and most capable model. Thinking is always on, so part of every reply budget
# goes to thinking. It needs an account on 30-day data retention; a zero-retention org gets a 400.
# ---------------------------------------------------------------
# STEP C. The three inputs, read from the data/ folder
# ---------------------------------------------------------------

DATA = Path(__file__).with_name("data")
EMAIL = (DATA / "padma_email.txt").read_text(encoding="utf-8")        # E1, E2, E3, E6
CLAIM_TEXT = (DATA / "refund_claim.txt").read_text(encoding="utf-8")  # E4 voting
BRIEF = (DATA / "brief_kf2481.txt").read_text(encoding="utf-8")       # E5, E8


# ---------------------------------------------------------------
# STEP D. The two tool descriptions. A schema is a phone-book entry:
#         name, what it does, what input it needs. It is NOT the code.
# ---------------------------------------------------------------
TOOLS_KAVERI = [
    {
        "name": "get_shipment_status",
        "description": "Live status for one Kaveri consignment id (e.g. KF-2481).",
        "input_schema": {
            "type": "object",
            "properties": {"consignment_id": {"type": "string"}},
            "required": ["consignment_id"],
        },
```

```
    },
    {
        "name": "find_alternate_carriers",
        "description": "Carriers that can take a lane (e.g. Pune-Chennai) within 48 hours.",
        "input_schema": {
            "type": "object",
            "properties": {"lane": {"type": "string"}},
            "required": ["lane"],
        },
    },
]
# ---------------------------------------------------------------
# STEP E. The tools themselves. Plain Python. The model never sees this code.
#         In production these would query the depot database or the MCP server.
# ---------------------------------------------------------------

def fake_status(consignment_id):
    return {"id": consignment_id, "status": "CUSTOMS_HOLD", "missing": "e-way bill", "days_late": 3}


def fake_carriers(lane):
    return ["BlueDart", "Delhivery"]
# ---------------------------------------------------------------
# STEP F. Helpers every exhibit uses
# ---------------------------------------------------------------

def text_of(reply):
    """Collect the text blocks only. Never read content[0]: a thinking block may come first."""
    answer = ""
    for block in reply.content:
        if block.type == "text":
            answer = answer + block.text
    return answer
def trace(turn, reply):
    """Print one line per call: the turn number, the stop_reason, and any tools the model asked for."""
    tool_names = []
    for block in reply.content:
        if block.type == "tool_use":
            tool_names.append(block.name)
    line = "# turn " + str(turn) + "  stop_reason=" + str(reply.stop_reason)
    if tool_names:
        line = line + "  (" + ", ".join(tool_names) + ")"
    print(line)
def _sdk_kwargs():
    """An org-level key must name a workspace on every request. Set ANTHROPIC_WORKSPACE_ID in .env,
    or use a key created inside a workspace, in which case this header is not needed."""
    workspace = os.environ.get("ANTHROPIC_WORKSPACE_ID")
    if workspace:
        return {"default_headers": {"anthropic-workspace-id": workspace}}
    return {}


def make_client():
    """The stand-in on battery, the real SDK live. The exhibits do not know or care which one they got."""
    if DRY_RUN:
        from standin import FakeClient
        return FakeClient()
    import anthropic                                    # only imported on the paid path
    # An explicit timeout (seconds) is required: with max_tokens this high the SDK would otherwise refuse a
    # non-streaming call ("Streaming is strongly recommended"). Ten minutes is far more than any exhibit needs.
    return anthropic.Anthropic(timeout=600.0, **_sdk_kwargs())
FALLBACK = {"betas": ["server-side-fallback-2026-07-01"], "fallbacks": "default"}
# Fable 5.1 runs safety classifiers that can decline an ordinary request (stop_reason "refusal").
# fallbacks="default" asks the API to re-run a declined request on the model Anthropic recommends for that
# category, inside the same call, instead of the exhibit crashing. The stand-in has no beta surface.


def create_message(**arguments):
    """messages.create with Fable's safety net (see FALLBACK above). Every ask*() helper goes through here."""
    client = make_client()
    if DRY_RUN:
        return client.messages.create(**arguments)
    return client.beta.messages.create(**FALLBACK, **arguments)
def ask(model, question, system=None, max_tokens=MAX_TOKENS):
    """One call, text in, text out. No tools. Checks stop_reason before trusting the text."""
    arguments = {
        "model": model,
        "max_tokens": max_tokens,
        "messages": [{"role": "user", "content": question}],
```

```
    }
    if system is not None:
        arguments["system"] = system

    reply = create_message(**arguments)
    if reply.stop_reason == "refusal":
        raise RuntimeError("The model declined to answer.")
    if reply.stop_reason == "max_tokens":
        # The reply was cut off. On current models the thinking block comes first and counts toward
        # max_tokens, so a cut-off reply can hold NO text at all. Fail loudly rather than return "".
        raise RuntimeError("Reply cut off at max_tokens=" + str(max_tokens) + " before it finished. Raise
            MAX_TOKENS.")
    return text_of(reply).strip()


def ask_json(model, question, max_tokens=MAX_TOKENS):
    """ask(), then parse the reply as JSON. Models sometimes wrap JSON in a ```json fence; strip it."""
    raw = ask(model, question, max_tokens=max_tokens)
    if raw.startswith("```"):
        lines = raw.split("\n")
        lines = lines[1:]                       # drop the opening ```json line
        if lines and lines[-1].strip() == "```":
            lines = lines[:-1]                  # drop the closing ``` line
        raw = "\n".join(lines)
    return json.loads(raw)
async def ask_async(model, question, max_tokens=MAX_TOKENS):
    """The async twin of ask(), for E4. On battery it sleeps 0.2 s so the slowest-scout lesson is visible."""
    if DRY_RUN:
        from standin import canned_text
        await asyncio.sleep(0.2)
        return canned_text(None, question)

    import anthropic
    client = anthropic.AsyncAnthropic(timeout=600.0, **_sdk_kwargs())
    reply = await client.beta.messages.create(
        model=model,
        max_tokens=max_tokens,
        messages=[{"role": "user", "content": question}],
        **FALLBACK,
    )
    if reply.stop_reason == "refusal":
        raise RuntimeError("The model declined to answer.")
    if reply.stop_reason == "max_tokens":
        raise RuntimeError("Reply cut off at max_tokens=" + str(max_tokens) + " before it finished. Raise
            MAX_TOKENS.")
    return text_of(reply).strip()
```

# E1 The augmented LLM: the code

One cell. One call, one tool, at most one round-trip.

```
# E1 · The augmented LLM: KAVI triages one exception  (objective 1.3a, the cell every organism is built from)
#
# One call, one tool, at most one round-trip. The model may ask us to look the shipment up.
# If it does, we run the lookup and send the fact back. Then it answers.
#
# Run on battery:  DRY_RUN=1 python e1_augmented_llm.py      (Windows:  set DRY_RUN=1  first)
# Run live:        python e1_augmented_llm.py                (needs ANTHROPIC_API_KEY, see common.py)

import json
from common import make_client, trace, text_of, fake_status, MODEL, MAX_TOKENS, EMAIL, TOOLS_KAVERI, DRY_RUN


# ---------------------------------------------------------------
# STEP 0. Things we decide before talking to the model
#         (MODEL, MAX_TOKENS and EMAIL come from common.py)
# ---------------------------------------------------------------
# The ONE tool the model is allowed to ask for. This is only a description: name, what it does,
# what input it needs. The model cannot run it. We run it.
TOOLS = [TOOLS_KAVERI[0]]                # get_shipment_status

# The instructions taped to the model's desk.
SYSTEM = (
    "You triage Kaveri shipment exceptions. "
    "Reply as JSON {category, urgency} and nothing else. "
    "Categories: delay, damage, customs, refund, address, other."
)

# STEP 1. The tool itself lives in common.py as fake_status(): plain Python. The model never sees it.
def triage(email_text):

    # STEP 2. Connect. DRY_RUN=1 gives the scripted stand-in; otherwise the real SDK.
    client = make_client()

    # STEP 3. Start the conversation with one message: the email. This is page 1 of the case file.
    conversation = []
    conversation.append({"role": "user", "content": email_text})

    # STEP 4. First call to the model.
    reply = client.messages.create(
        model=MODEL,
        max_tokens=MAX_TOKENS,
        system=SYSTEM,
        tools=TOOLS,
        messages=conversation,
    )
    trace(1, reply)                      # prints:  # turn 1  stop_reason=...  (tool names)
    # STEP 5. Did the model ask for the tool?  "tool_use" means: "I need a lookup before I can answer."
    if reply.stop_reason == "tool_use":

        # 5a. Find the tool request inside the reply. A reply is a list of blocks.
        tool_request = None
        for block in reply.content:
            if block.type == "tool_use":
                tool_request = block
                break

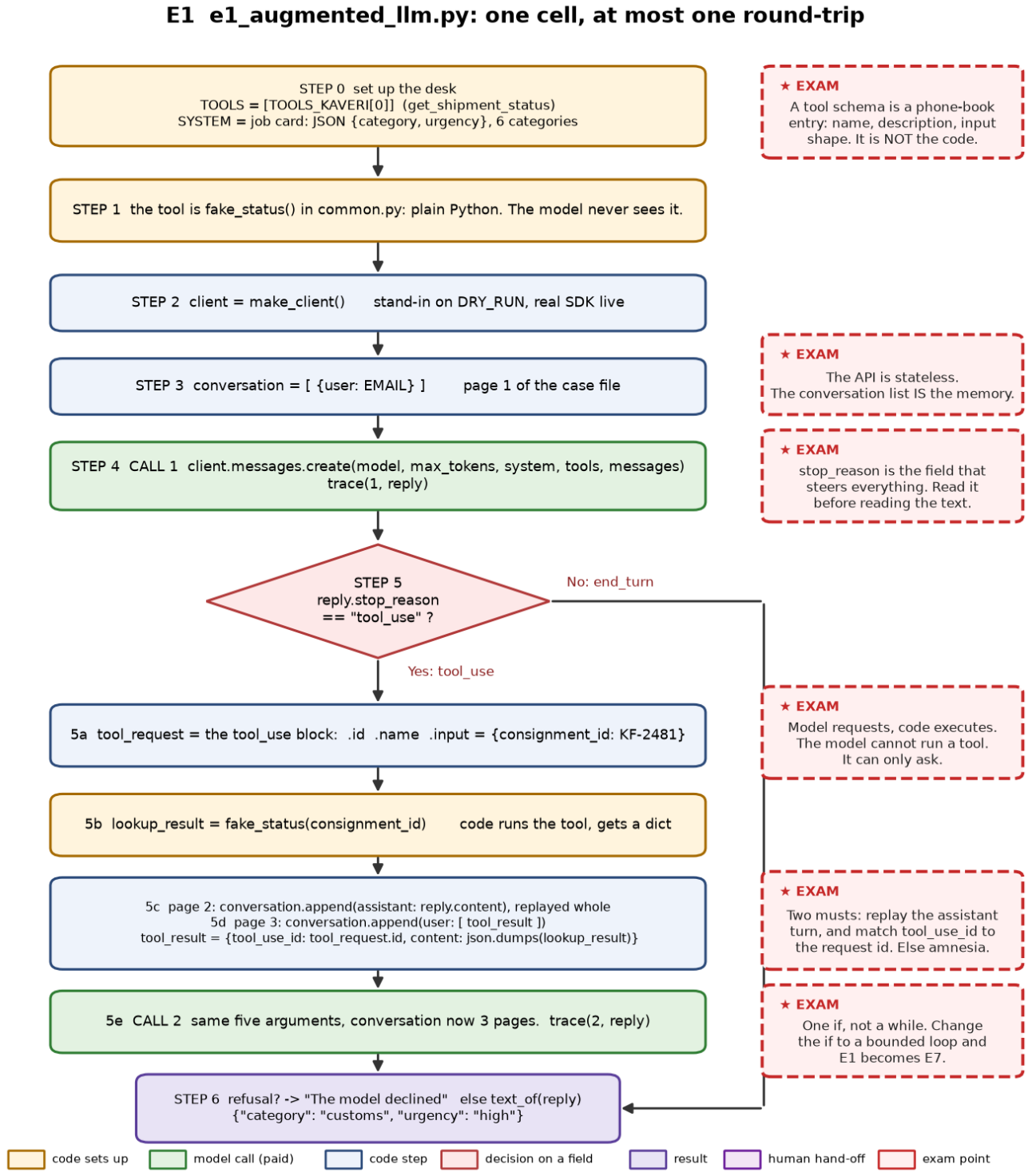
        # 5b. Run the tool ourselves, with the input the model filled in.
        consignment_id = tool_request.input["consignment_id"]
        lookup_result = fake_status(consignment_id)
        # 5c. Add the model's own request to the conversation: page 2.
        #     The API remembers nothing between calls, so we must send its own words back.
        conversation.append({"role": "assistant", "content": reply.content})

        # 5d. Add the tool's answer: page 3. It is tagged with the request's id so the model knows
        #     which question this answers. The API wants text, so the dict becomes a JSON string.
        tool_answer = {
            "type": "tool_result",
            "tool_use_id": tool_request.id,
            "content": json.dumps(lookup_result),
        }
        conversation.append({"role": "user", "content": [tool_answer]})
        # 5e. Second call. Same five arguments. The conversation is now 3 pages long.
        reply = client.messages.create(
            model=MODEL,
```

```
            max_tokens=MAX_TOKENS,
            system=SYSTEM,
            tools=TOOLS,
            messages=conversation,
        )
        trace(2, reply)

    # STEP 6. Read the final answer. A refusal is not an answer; otherwise collect the text blocks only.
    if reply.stop_reason == "refusal":
        return "The model declined to answer."
    return text_of(reply)
if __name__ == "__main__":
    if DRY_RUN:
        print("# E1 · augmented LLM · DRY_RUN: scripted stand-in, 0 paid calls")
    else:
        print("# E1 · augmented LLM · LIVE · " + MODEL)
    print(triage(EMAIL))
```

# E1 The augmented LLM: the flowchart




## Exam points on this chart

★ A tool schema is a description, not code. The model requests; your code executes.
★ The API is stateless: the conversation list is the memory. Replay the assistant turn and match tool_use_id.
★ stop_reason steers everything. E1 uses one if; E7 turns that if into a bounded loop.

# E2 Prompt chaining: the code

Two desks in a fixed order with a plain-Python gate between them.

```
# E2 · Prompt chaining: two desks in a fixed order, a checklist between them  (objective 1.3b, workflow 1)
#
#   Desk 1: read the email, say ONE word: what kind of problem is this?
#   Gate:   plain Python checks that word is on the approved list. If not, stop the line.
#   Desk 2: knowing the category, write a short courteous reply.
# The CODE decides the order. The model never gets to change it. That is Law 2.
#
# Run on battery:  DRY_RUN=1 python e2_chaining.py          Run live:  python e2_chaining.py

from common import ask, MODEL, EMAIL, DRY_RUN


# ------------------------------------------------------------
# STEP 0. The laminated checklist on the wall between the two desks.
#         Only these six words are allowed to pass from desk 1 to desk 2.
# ------------------------------------------------------------
VALID_CATEGORIES = {"delay", "damage", "customs", "refund", "address", "other"}


# ------------------------------------------------------------
# STEP 1. DESK 1: classify. One word only, on the small model.
# ------------------------------------------------------------

def classify(email_text):
    question = (
        "Read this customer email and reply with ONE word only, the category of the problem. "
        "Choose from: delay, damage, customs, refund, address, other. "
        "No explanation, no punctuation, just the word.\n\n"
        + email_text
    )
    word = ask(MODEL, question)
    return word.lower()                  # "Customs" and "customs" become the same thing
# ------------------------------------------------------------
# STEP 2. THE GATE: plain Python. No model, no cost.
#         If desk 1 mumbled something that is not on the checklist, stop the line here.
# ------------------------------------------------------------

def gate(category):
    if category not in VALID_CATEGORIES:
        raise ValueError("bad category: " + category)
    return category


# ------------------------------------------------------------
# STEP 3. DESK 2: draft the reply. It is TOLD the category; it does not re-decide it.
# ------------------------------------------------------------
def draft_reply(email_text, category):
    question = (
        "This is a " + category + " exception at Kaveri Freight. "
        "Draft a courteous reply to the customer, under 120 words. "
        "Do not promise any refund or compensation.\n\n"
        + email_text
    )
    return ask(MODEL, question)

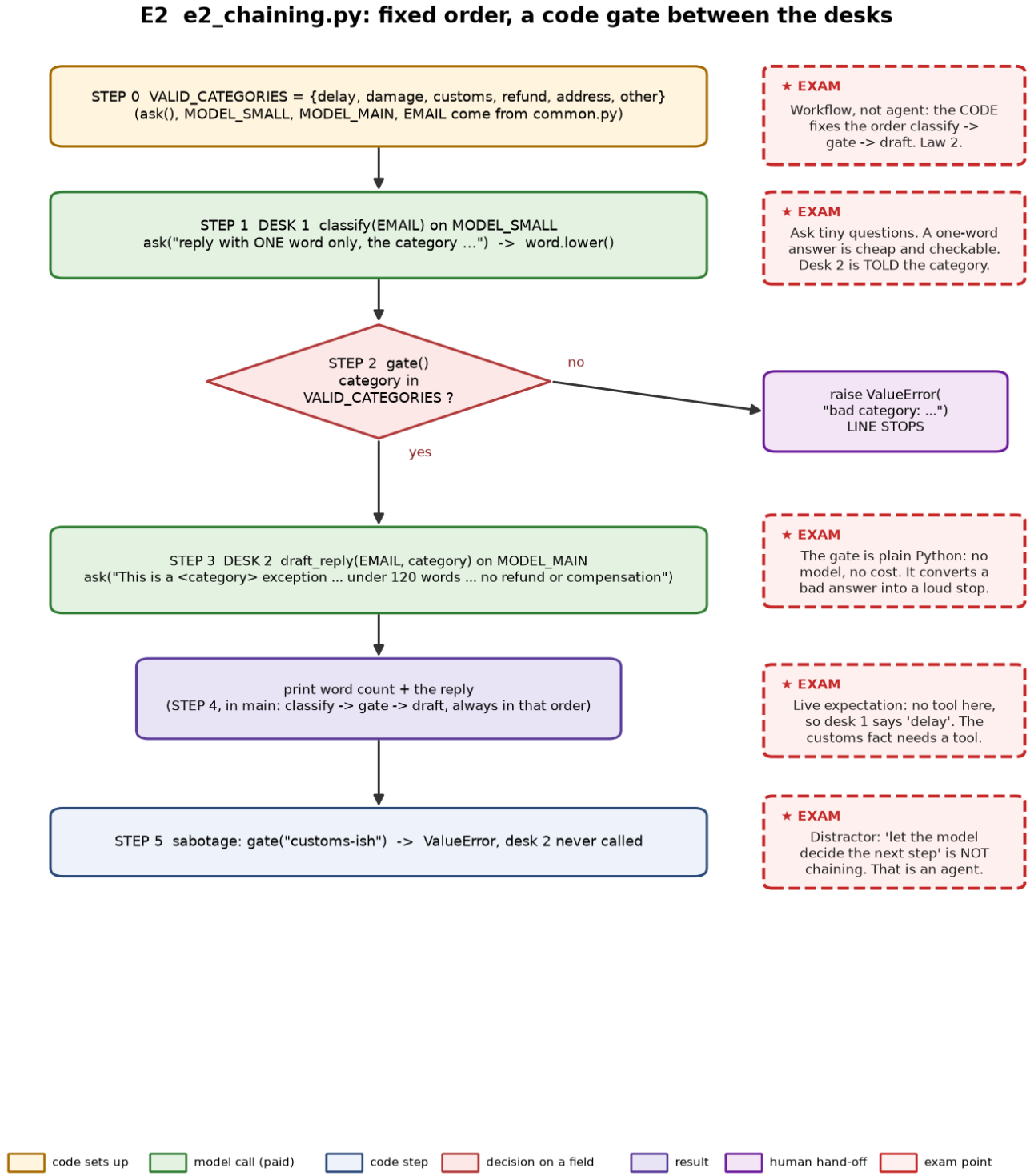

if __name__ == "__main__":
    print("# E2 · prompt chaining · " + ("DRY_RUN" if DRY_RUN else "LIVE"))
    # STEP 4. Run the line in the order the code fixes: classify -> gate -> draft.
    print("=== Desk 1: classify ===")
    category = classify(EMAIL)
    print("Desk 1 said:", repr(category))

    print()
    print("=== Gate: check the word against the list ===")
    category = gate(category)
    print("Gate passed:", category, "is on the approved list. Line continues.")

    print()
    print("=== Desk 2: draft ===")
    reply = draft_reply(EMAIL, category)
    print("Desk 2 wrote", len(reply.split()), "words:")
    print()
    print(reply)
```

```
# STEP 5. Sabotage test: hand the gate a word that is not on the list. No model call here.
print()
print("=== Sabotage: feed the gate a word that is not on the list ===")
try:
    gate("customs-ish")
except ValueError as error:
    print("Gate stopped the line:", error)
    print("Desk 2 was never called. No reply was written for a nonsense category.")
```

# E2 Prompt chaining: the flowchart




### Exam points on this chart

★ Workflow, not agent: the code fixes the order. Law 2.
★ Ask tiny questions; a one-word answer is cheap and checkable. Desk 2 is told the category.
★ The gate costs nothing and converts a bad answer into a loud stop.

# E3 Routing: the code

A cheap nurse call picks the ward; the ward picks the model.

```
# E3 · Routing: a cheap nurse at the door, specialist wards behind it  (objective 1.3b, workflow 2)
#
#   Nurse:  one cheap call says ONE word: which kind of problem is this?   (E2's classify + gate, reused)
#   Table:  the word picks WHICH specialist, and which MODEL, handles the email.
#   Ward:   ONE call to that specialist. It never saw a menu. The code chose.
# Routing is where model selection stops being one decision and becomes a decision PER category.
#
# Run on battery:  DRY_RUN=1 python e3_routing.py           Run live:  python e3_routing.py

from common import ask, MODEL, EMAIL, DRY_RUN
from e2_chaining import classify, gate          # reuse E2's nurse and checklist
# ------------------------------------------------------------
# STEP 0. The wards. Each category maps to (which model, the ward's instructions).
#         This table IS the router. Add a ward by adding a row. No model is involved in the choice.
# ------------------------------------------------------------

SPECIALISTS = {
    "refund":  (MODEL,  "You resolve refund disputes at Kaveri Freight. Acknowledge, say what you will verify,
        never promise money."),
    "customs": (MODEL,  "You explain customs holds and next steps for Indian import consignments at Kaveri
        Freight."),
    "delay":   (MODEL, "You write short delay apologies for Kaveri Freight with a new ETA. Under 80 words."),
}
# One MODEL for every ward here. Production would give each ward its own tier: refunds on the strongest,
    apologies on the cheapest.
# If the nurse says a word that has no ward (for example "address"), send it here instead of crashing.
DEFAULT_WARD = "delay"


# ------------------------------------------------------------
# STEP 1. Table lookup: plain Python. An unmapped word goes to a safe ward, not a KeyError at 2 a.m.
# ------------------------------------------------------------

def pick_ward(category):
    if category in SPECIALISTS:
        return category
    return DEFAULT_WARD
# ------------------------------------------------------------
# STEP 2. The router: email in, specialist's reply out.
# ------------------------------------------------------------

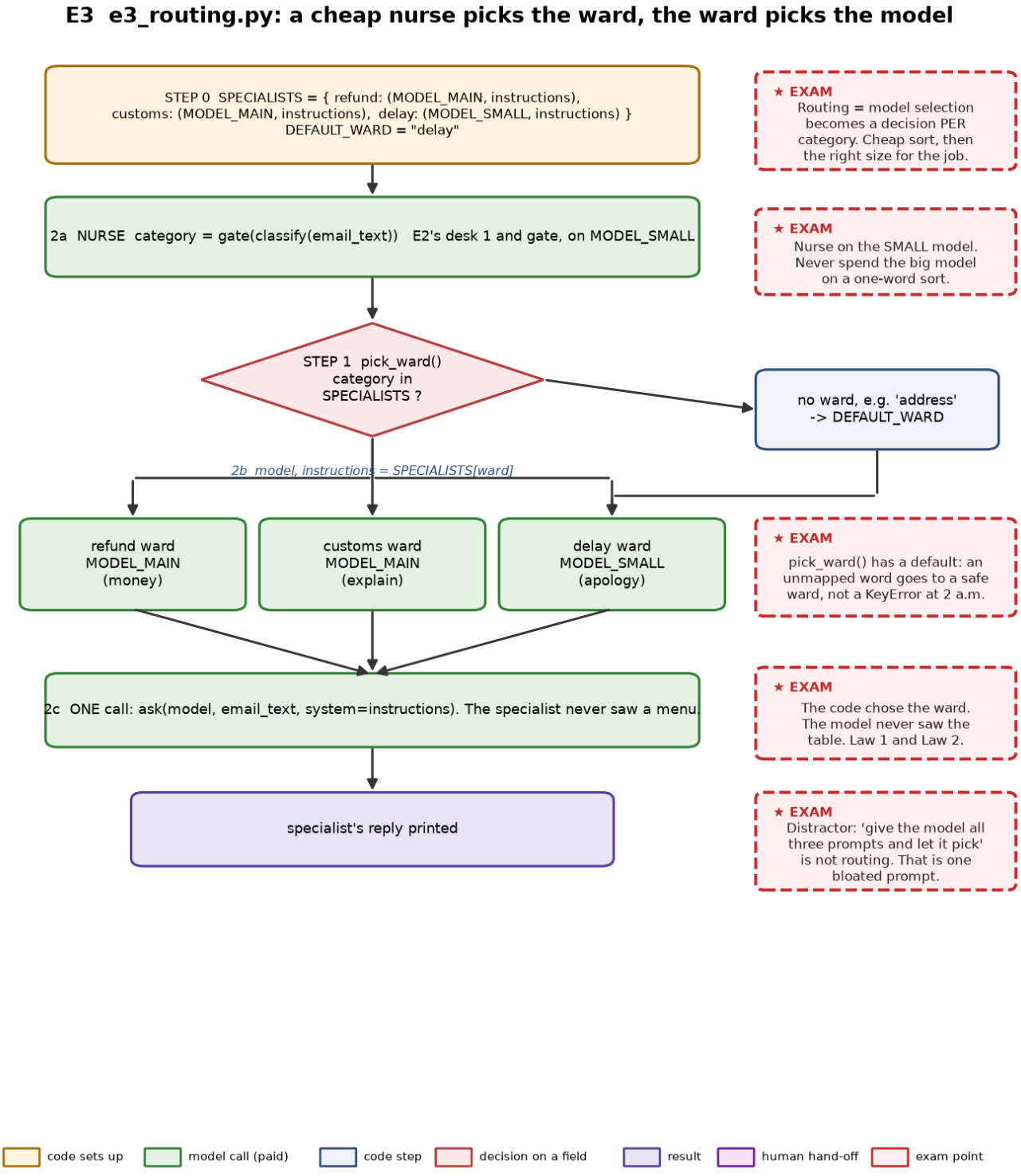
def route(email_text):
    # 2a. The nurse looks once and says one word. The gate checks it.
    category = gate(classify(email_text))

    # 2b. The word picks the ward; the ward brings its own model and instructions.
    ward = pick_ward(category)
    model, instructions = SPECIALISTS[ward]
    print("Nurse said:", repr(category), "-> ward:", ward, "-> model:", model)
    # 2c. ONE call to the chosen specialist. Its instructions go in the system prompt.
    return ask(model, email_text, system=instructions)


if __name__ == "__main__":
    print("# E3 · routing · " + ("DRY_RUN" if DRY_RUN else "LIVE"))

    # STEP 3. Route Padma's email.
    print("=== Routing Padma's email ===")
    specialist_reply = route(EMAIL)
    print()
    print("Specialist wrote:")
    print(specialist_reply)
    # STEP 4. Show the default ward. No model call: just the table lookup.
    print()
    print("=== A word with no ward ===")
    word = "address"
    ward = pick_ward(word)
    print(repr(word), "has no ward, so it goes to", repr(ward), "on model", SPECIALISTS[ward][0])
```

# E3 Routing: the flowchart




## Exam points on this chart

★ Routing is where model selection becomes a decision per category.
★ Nurse on the small model; never spend the big model on a one-word sort.
★ pick_ward() has a default: an unmapped word is a safe fallback, not a KeyError.

# E4 Parallelization: the code

Sectioning for coverage, voting for confidence. Same trick, opposite purposes.

```
# E4 · Parallelization: three scouts, or three judges  (objective 1.3b, workflow 3)
#
#   Sectioning: three DIFFERENT questions at once (three scouts on three roads). Purpose: coverage, speed.
#   Voting:     the SAME question three times (three judges). Purpose: confidence. Costs 3x.
# Same trick, simultaneity, opposite purposes. The CODE drew the fork; no model chose to parallelize.
#
# Run on battery:  DRY_RUN=1 python e4_parallel.py   (each call sleeps 0.2 s)      Run live:  python
    e4_parallel.py

import asyncio
import time
from common import ask_async, MODEL, CLAIM_TEXT, DRY_RUN
# ------------------------------------------------------------
# STEP 0. Things we decide before talking to the model
#         (ask_async in common.py uses the async SDK client, so several calls can be in flight at once)
# ------------------------------------------------------------

CONSIGNMENT = "KF-2481"
CARRIERS = ["BlueDart", "Delhivery", "IndiaPost"]


# ------------------------------------------------------------
# STEP 1. SECTIONING: three different questions, all at once
# ------------------------------------------------------------
async def check_one_carrier(carrier):
    question = (
        "In one line, give a plausible ETA for consignment " + CONSIGNMENT
        + " on the Pune-Chennai lane if re-routed via " + carrier + ". "
        "Assume it is currently held at customs. Start your line with the carrier name."
    )
    return await ask_async(MODEL, question)


async def sectioning():
    # asyncio.gather starts all three and waits for the slowest. Wall clock = the slowest scout.
    return await asyncio.gather(
        check_one_carrier(CARRIERS[0]),
        check_one_carrier(CARRIERS[1]),
        check_one_carrier(CARRIERS[2]),
    )
async def sequential():
    # The same three questions one after another, for comparison. Wall clock = the sum.
    results = []
    for carrier in CARRIERS:
        results.append(await check_one_carrier(carrier))
    return results


# ------------------------------------------------------------
# STEP 2. VOTING: the same question three times, majority wins
# ------------------------------------------------------------
async def one_vote():
    question = (
        "You are a claims reviewer at Kaveri Freight. Read this refund claim and reply with "
        "exactly one word, APPROVE or REJECT. No explanation.\n\n" + CLAIM_TEXT
    )
    return await ask_async(MODEL, question)

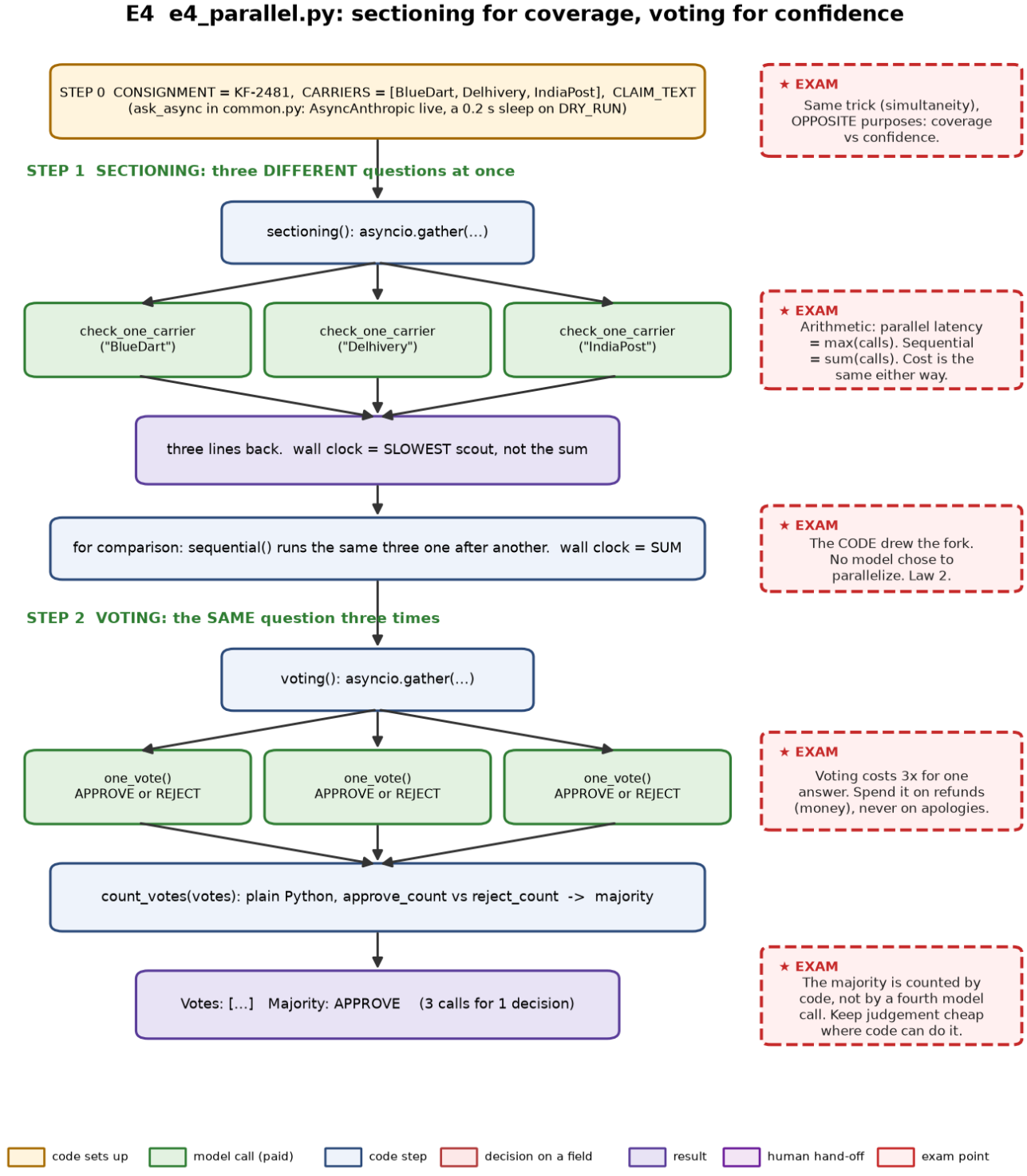

def count_votes(votes):
    # Plain Python counts. No fourth model call to decide the majority.
    approve_count = 0
    reject_count = 0
    for vote in votes:
        if "APPROVE" in vote.upper():
            approve_count = approve_count + 1
        else:
            reject_count = reject_count + 1
    if approve_count > reject_count:
        return "APPROVE"
    return "REJECT"
async def voting():
    votes = await asyncio.gather(one_vote(), one_vote(), one_vote())
    return votes, count_votes(votes)
```

```
# ---------------------------------------------------------------
# STEP 3. Run both, and time the sectioning against the sequential version
# ---------------------------------------------------------------

async def main():
    print("=== Sectioning: three scouts at once ===")
    start = time.perf_counter()
    scouts = await sectioning()
    parallel_seconds = time.perf_counter() - start
    for line in scouts:
        print("  scout ->", line)
    print()
    print("=== The same three scouts, one after another ===")
    start = time.perf_counter()
    await sequential()
    sequential_seconds = time.perf_counter() - start
    print("Parallel took", round(parallel_seconds, 2), "s (about the slowest single call).")
    print("Sequential took", round(sequential_seconds, 2), "s (about the sum of all three).")

    print()
    print("=== Voting: three judges on one refund claim ===")
    votes, majority = await voting()
    print("Votes:", list(votes))
    print("Majority:", majority)
    print("Note: three calls for one decision. Spend that on refunds, not on apologies.")
if __name__ == "__main__":
    print("# E4 · parallelization · " + ("DRY_RUN (each call sleeps 0.2 s)" if DRY_RUN else "LIVE"))
    asyncio.run(main())
```

# E4 Parallelization: the flowchart




## Exam points on this chart

★ Parallel latency = max(calls); sequential = sum(calls). Cost is the same either way.
★ Voting costs 3x for one decision: spend it on money, not on apologies.
★ The code drew the fork and the code counts the votes.

# E5 Orchestrator-workers: the code

The foreman model writes the plan; code steers the workers; synthesis writes one recommendation.

```
# E5 · Orchestrator-workers: the site foreman writes the plan himself  (objective 1.5, and workflow 4 of 1.3b)
#
#   Foreman (model):   reads the brief, writes a list of subtasks as JSON.
#   Workers (model):   one call per subtask. Each sees ONLY the findings it declared it needs, plus the depot
    facts.
#   Synthesis (model): sees all findings, writes ONE recommendation.
# The SHAPE (plan -> work -> synthesise) is fixed code. The CONTENT of the plan is the model's.
#
# Run on battery:  DRY_RUN=1 python e5_orchestrator_workers.py      Run live:  python e5_orchestrator_workers.py

import json
import textwrap
from common import ask, ask_json, fake_status, fake_carriers, MODEL, BRIEF, DRY_RUN
WIDTH = 100                                          # wrap long model text so the terminal stays readable


# ---------------------------------------------------------------
# STEP 0. The facts the depot system can supply. The CODE hands these to the workers.
#         Workers reason over facts; they do not invent them.
# ---------------------------------------------------------------

FACTS = {
    "shipment": fake_status("KF-2481"),
    "carriers_on_lane": fake_carriers("Pune-Chennai"),
}
# ---------------------------------------------------------------
# STEP 1. FOREMAN: the model writes the plan
# ---------------------------------------------------------------

def make_plan(brief):
    question = (
        "Brief:\n" + brief + "\n\n"
        "Break this into 2 to 5 subtasks. Reply with JSON only, a list like:\n"
        '[{"id": 1, "task": "status check", "needs": []}, {"id": 2, "task": "carrier options", "needs": [1]}]\n'
        '"needs" lists the ids of earlier subtasks whose findings this one depends on.'
    )
    return ask_json(MODEL, question)
# ---------------------------------------------------------------
# STEP 2. WORKER: one subtask, on the small model.
#         It is shown ONLY the findings it declared it needs (context isolation), plus the depot facts.
# ---------------------------------------------------------------

def run_worker(subtask, findings_so_far):
    known = {}
    for needed_id in subtask["needs"]:
        if needed_id in findings_so_far:
            known[needed_id] = findings_so_far[needed_id]

    question = (
        "You are a worker at Kaveri Freight. Do this one subtask and reply in one or two lines.\n"
        "Subtask: " + subtask["task"] + "\n"
        "Known findings from earlier subtasks: " + json.dumps(known) + "\n"
        "Depot facts you may use: " + json.dumps(FACTS)
    )
    return ask(MODEL, question)
# ---------------------------------------------------------------
# STEP 3. SYNTHESIS: sees everything, writes ONE recommendation
# ---------------------------------------------------------------

def synthesise(brief, findings):
    question = (
        "Brief:\n" + brief + "\n\n"
        "Findings from the workers:\n" + json.dumps(findings, indent=2) + "\n\n"
        "Write ONE recommendation for Meera's desk, three sentences at most, starting with RECOMMENDATION:"
    )
    return ask(MODEL, question)
# ---------------------------------------------------------------
# STEP 4. Printing helpers. Nothing here touches the model: they only lay the output out.
# ---------------------------------------------------------------

def heading(text):
    print()
    print("=== " + text + " ===")
def block(label, text, indent="    "):
```

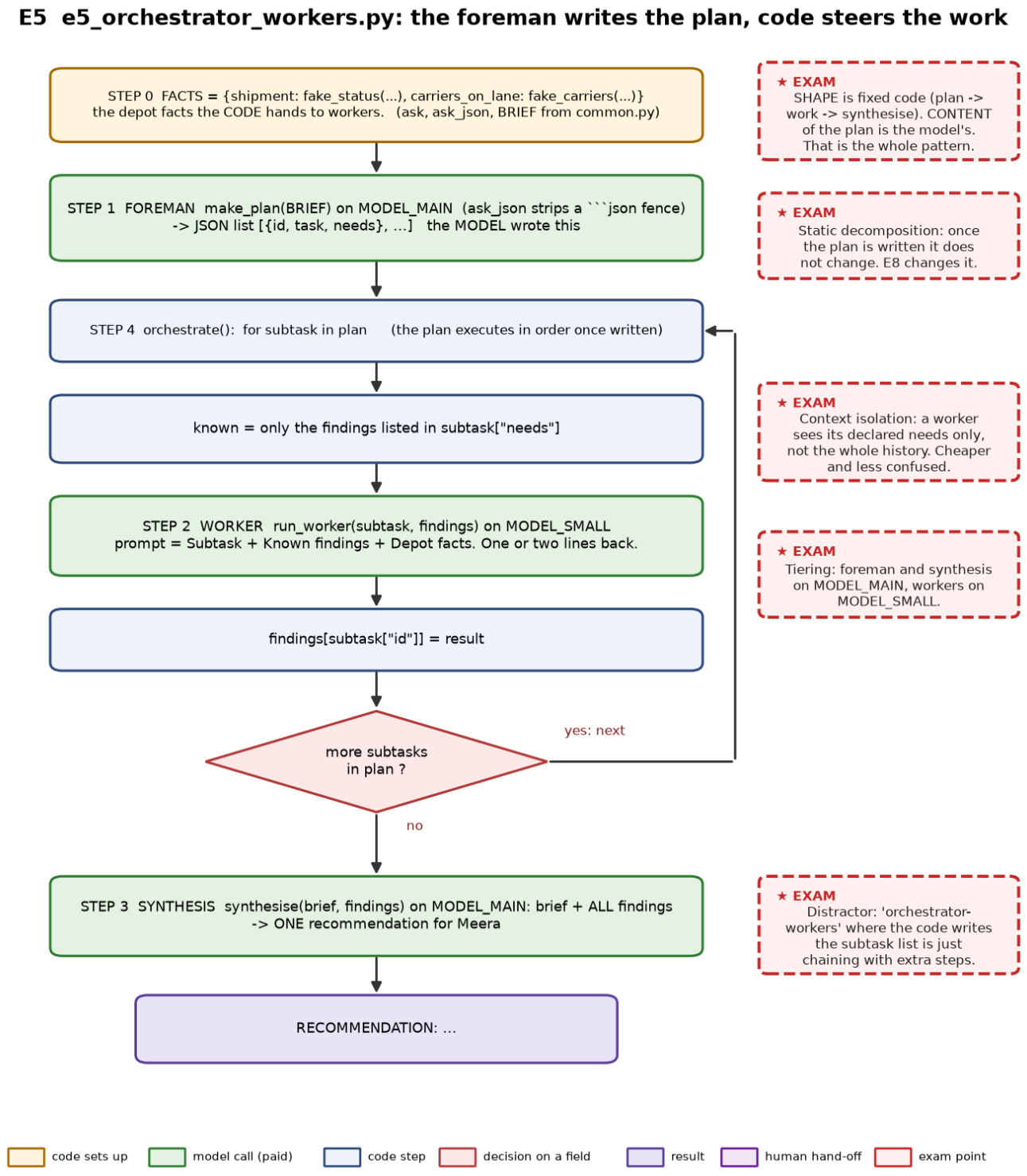
```
    """A label on its own line, then the text wrapped underneath it."""
    print(label)
    for paragraph in text.strip().split("\n"):
        print(textwrap.fill(paragraph, WIDTH, initial_indent=indent, subsequent_indent=indent) if
            paragraph.strip() else "")
def show_plan(plan):
    """The foreman's plan as a table: id, what it waits for, the task."""
    print("  id  needs   task")
    for subtask in plan:
        needs = ",".join(str(n) for n in subtask["needs"]) or "-"
        first, *rest = textwrap.wrap(subtask["task"], WIDTH - 14) or [""]
        print("  " + str(subtask["id"]).ljust(4) + needs.ljust(8) + first)
        for line in rest:
            print(" " * 14 + line)


# ------------------------------------------------------------------
# STEP 5. The fixed shape: plan, then work in plan order, then synthesise
# ------------------------------------------------------------------
def orchestrate(brief):
    heading("FOREMAN · the model writes the plan (1 call)")
    plan = make_plan(brief)
    show_plan(plan)

    heading("WORKERS · one call per subtask, each sees only the findings it asked for")
    findings = {}
    for subtask in plan:                                    # the plan executes in order once written
        result = run_worker(subtask, findings)
        findings[subtask["id"]] = result
        print()
        block("Worker " + str(subtask["id"]) + " · needs " + (str(subtask["needs"]) if subtask["needs"] else
            "nothing")
              + " · " + textwrap.shorten(subtask["task"], 70), result)
    heading("SYNTHESIS · sees every finding, writes one recommendation (1 call)")
    recommendation = synthesise(brief, findings)
    block("", recommendation, indent="  ")
    print()
    print("Calls made: 1 plan + " + str(len(plan)) + " workers + 1 synthesis = " + str(len(plan) + 2))
    return recommendation


if __name__ == "__main__":
    print("# E5 · orchestrator-workers · " + ("DRY_RUN" if DRY_RUN else "LIVE"))
    orchestrate(BRIEF)
```

# E5 Orchestrator-workers: the flowchart




### Exam points on this chart

★ Shape is fixed code; content of the plan is the model's. Static decomposition.
★ Context isolation: a worker sees only its declared needs, plus facts the code supplies.
★ Tiering: one MODEL here; production would put workers on a cheaper tier than the foreman.

# E6 Evaluator-optimizer: the code

Writer and editor, with a page limit.

```
# E6 · Evaluator-optimizer: a writer and an editor, with a page limit  (objective 1.3b, workflow 5)
#
#   Writer (model):  drafts a reply.
#   Editor (model):  checks it against a rubric. Says PASS, or one line of criticism.
#   Writer again:    revises using the criticism. Repeat.
# Two organs: MAX_ROUNDS is the budget, PASS is the exit. E7's agent loop has exactly these two, grown larger.
#
# Run on battery:  DRY_RUN=1 python e6_evaluator_optimizer.py      Run live:  python e6_evaluator_optimizer.py

from common import ask, MODEL, EMAIL, DRY_RUN


# ------------------------------------------------------------
# STEP 0. The budget and the rubric
# ------------------------------------------------------------
MAX_ROUNDS = 2                          # the budget: the editor gets at most two looks

RUBRIC = (
    "You are an editor at Kaveri Freight. Check the draft below against this rubric:\n"
    "  1. Tone courteous?\n"
    "  2. Facts only, no promises of refunds or compensation?\n"
    "  3. Under 120 words?\n"
    "If all three pass, reply with exactly the single word PASS. "
    "Otherwise reply with ONE line of specific criticism and nothing else.\n\n"
    "Draft:\n"
)
# ------------------------------------------------------------
# STEP 1. The loop: draft once, then (evaluate, maybe revise) up to MAX_ROUNDS times
# ------------------------------------------------------------

def draft_and_polish(email_text):

    # 1a. WRITER, first draft. Loose instructions on purpose, so the editor has something to catch.
    draft = ask(MODEL, "Reply to this customer exception email on behalf of Kaveri Freight:\n\n" + email_text)
    print("Draft 0:", len(draft.split()), "words")
    if not draft:
        return "ESCALATED to Meera's desk: the writer returned an empty draft."   # the floor: never ship
            nothing
    # 1b. The editing rounds. range(1, MAX_ROUNDS + 1) is the budget.
    for round_number in range(1, MAX_ROUNDS + 1):

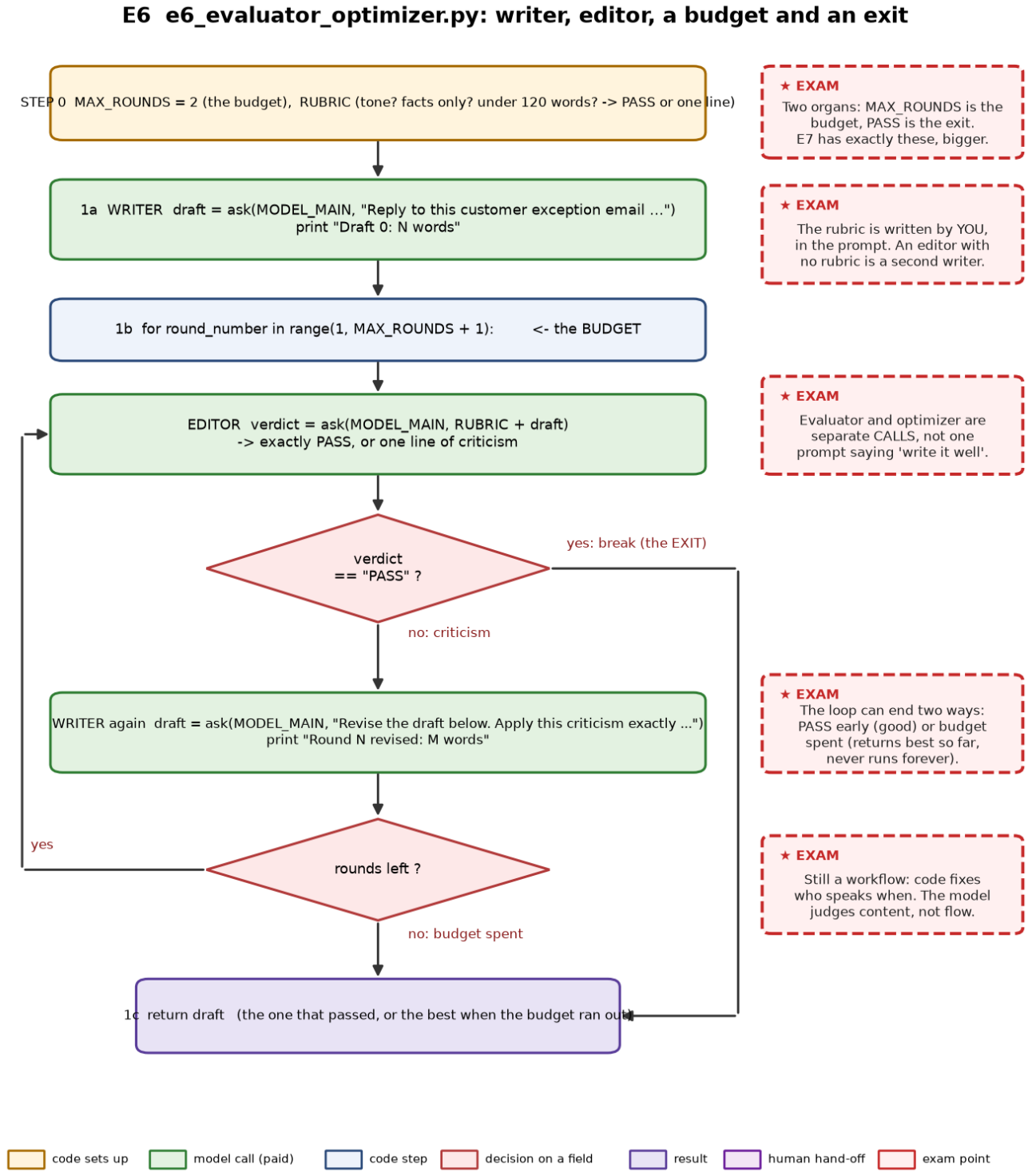
        # EDITOR: PASS, or one line of criticism.
        verdict = ask(MODEL, RUBRIC + draft)
        print("Round", round_number, "editor said:", verdict)

        # The exit. PASS means stop early and do not spend the remaining budget.
        if verdict == "PASS":
            break

        # WRITER again: revise using the criticism.
        draft = ask(MODEL,
                    "Revise the draft below. Apply this criticism exactly and change nothing else:\n"
                    + verdict + "\n\nDraft:\n" + draft)
        print("Round", round_number, "revised:", len(draft.split()), "words")
        if not draft:
            return "ESCALATED to Meera's desk: the writer returned an empty revision in round " +
                str(round_number) + "."
    # 1c. Whatever draft we hold now ships: the one that passed, or the best after the budget ran out.
    return draft


if __name__ == "__main__":
    print("# E6 · evaluator-optimizer · " + ("DRY_RUN" if DRY_RUN else "LIVE"))
    final = draft_and_polish(EMAIL)
    print()
    print("=== Final draft ===")
    print(final)
```

# E6 Evaluator-optimizer: the flowchart




### Exam points on this chart

★ Two organs: MAX_ROUNDS is the budget, PASS is the exit. E7 has exactly these, grown larger.
★ The rubric is yours and lives in the prompt. Evaluator and optimizer are separate calls.
★ The loop ends two ways: PASS early, or budget spent with the best draft so far.

# E7 The agent loop: the code

A bounded loop steered by stop_reason, every value on its own branch.

```
# E7 · The agent loop: every stop_reason lands somewhere explicit  (objective 1.3c, the 50 percent row)
#
# A loop around messages.create that keeps going while the model asks for tools and stops when it does not.
# The field that steers it is stop_reason. BUDGET is written before the loop exists. Escalation is the floor.
#
# This version NARRATES itself. For every turn it prints: the content blocks that came back, the stop_reason and
# the branch it chose, every tool call with its input and its result, how the case file grew, tokens and
    seconds.
# At the end: a summary of turns, tools, stop_reasons seen and the exit taken. The loop itself is unchanged.
#
# Run on battery:  DRY_RUN=1 python e7_agent_loop.py
#                  D1_SCENARIO=default|truncate|refuse|exhaust makes the stand-in exercise each branch
# Run live:        python e7_agent_loop.py
import json
import textwrap
import time
from common import make_client, text_of, fake_status, fake_carriers, MODEL, MAX_TOKENS, TOOLS_KAVERI, DRY_RUN,
    SCENARIO


# ---------------------------------------------------------------
# STEP 0. Things we decide BEFORE the loop exists
# ---------------------------------------------------------------

BUDGET = 6                                # the most turns we will ever pay for. Never "while True".
TOOLS = TOOLS_KAVERI                      # the detective's toolbag: both kaveri tools
BRIEF_E7 = "Why is KF-2481 stuck, and what are our options?"
WIDTH = 100                               # wrap width for printed text
# Every value stop_reason can take, what it means, and what THIS loop does about it. Printed as a legend
# before the loop runs, so the branches you do not see fire today are still in front of you.
STOP_REASONS = {
    "tool_use":    ("the model wants a tool run before it can go on",      "run every tool it asked for, send
        the results back, loop"),
    "end_turn":    ("the model finished its answer",                        "return the text: the natural
        exit"),
    "max_tokens":  ("the reply hit max_tokens before it finished",          "keep the partial text, append
        'Continue.', loop"),
    "pause_turn":  ("a server-side tool paused the turn",                   "send the reply back unchanged to
        resume, loop"),
    "refusal":     ("the model or a safety classifier declined",            "ESCALATE with the reason, never a
        blind retry"),
    "<other>":     ("stop_sequence, or a value that does not exist yet",    "ESCALATE with the value, never
        swallow it"),
}
# ---------------------------------------------------------------
# STEP 1. The tools, dispatched by name. Plain Python.
#         An unknown tool returns a structured error, not an exception: the model can read it and adapt.
# ---------------------------------------------------------------

def run_tool(tool_request):
    if tool_request.name == "get_shipment_status":
        return fake_status(tool_request.input["consignment_id"])
    if tool_request.name == "find_alternate_carriers":
        return fake_carriers(tool_request.input["lane"])
    return {"error": "unknown tool " + tool_request.name}
# ---------------------------------------------------------------
# STEP 2. The floor. When the loop cannot finish, a human gets the case WITH a reason.
# ---------------------------------------------------------------

def escalate(reason):
    return "ESCALATED to Meera's desk (" + reason + "): " + BRIEF_E7


# ---------------------------------------------------------------
# STEP 3. Narration helpers. They only print; the loop would run the same without them.
# ---------------------------------------------------------------
def say(label, text="", indent="      "):
    """A short label, then the text wrapped under it. Empty text prints the label alone."""
    print("    " + label)
    for paragraph in str(text).strip().split("\n"):
        if paragraph.strip():
            print(textwrap.fill(paragraph, WIDTH, initial_indent=indent, subsequent_indent=indent))
def describe_blocks(reply):
    """What came back in content, block by block. Never read content[0] blind: thinking may come first."""
```

```
    parts = []
    for block in reply.content:
        if block.type == "text":
            parts.append("text(" + str(len(block.text.split())) + " words)")
        elif block.type == "tool_use":
            parts.append("tool_use(" + block.name + ")")
        elif block.type == "thinking":
            shown = getattr(block, "thinking", "") or ""
            parts.append("thinking(" + ("hidden" if not shown else str(len(shown.split())) + " words") + ")")
        else:
            parts.append(block.type)
    return ", ".join(parts) if parts else "(no content blocks)"
def print_legend():
    print()
    print("=== THE SIX WAYS A TURN CAN END, AND WHERE EACH ONE LANDS ===")
    for value, (meaning, action) in STOP_REASONS.items():
        print("  " + value.ljust(12) + meaning.ljust(58) + "-> " + action)
    print()
    print("=== BEFORE THE LOOP: the things decided in code ===")
    print("  BUDGET      " + str(BUDGET) + " turns, then escalate. Never 'while True'.")
    print("  TOOLS       " + ", ".join(t["name"] for t in TOOLS) + "  (the model may ASK; the code RUNS)")
    print("  BRIEF       " + BRIEF_E7)
    print("  max_tokens  " + str(MAX_TOKENS) + "  (thinking counts toward it)")
def print_summary(log, exit_reason, final_text):
    print()
    print("=== SUMMARY ===")
    print("  turn  stop_reason  branch taken                 tools run                     tokens in/out
        seconds")
    for row in log:
        print("  " + str(row["turn"]).ljust(6) + row["stop"].ljust(13) + row["branch"].ljust(30)
              + (", ".join(row["tools"]) or "-").ljust(30) + (str(row["in"]) + "/" + str(row["out"])).ljust(16)
                  + str(row["secs"]))
    print()
    print("  turns used      : " + str(len(log)) + " of " + str(BUDGET))
    print("  tool calls      : " + str(sum(len(r["tools"]) for r in log)))
    print("  stop_reasons    : " + ", ".join(sorted(set(r["stop"] for r in log))) + "  (of the six in the
        legend)")
    print("  total tokens    : " + str(sum(r["in"] for r in log)) + " in, " + str(sum(r["out"] for r in log)) +
        " out")
    print("  exit            : " + exit_reason)
    print()
    print("=== WHAT THE DESK RECEIVES ===")
    print(textwrap.fill(final_text, WIDTH, initial_indent="  ", subsequent_indent="  ") if final_text else "
        (nothing)")
# ---------------------------------------------------------------
# STEP 4. The loop. Identical in logic to the plain version; the prints are the only addition.
# ---------------------------------------------------------------

def investigate():
    client = make_client()
    conversation = [{"role": "user", "content": BRIEF_E7}]  # the case file. The API is stateless, so this IS
        the memory.
    text_so_far = ""                                        # text that arrived before a max_tokens cut
    log = []                                                # one row per turn, for the summary

    print()
    print("=== THE LOOP ===")
    print("  case file starts with 1 message: the brief.")
    for turn in range(1, BUDGET + 1):                       # bounded, always
        print()
        print("--- turn " + str(turn) + " of " + str(BUDGET) + " · sending " + str(len(conversation)) + "
            message(s) and "
              + str(len(TOOLS)) + " tool schemas ---")
        started = time.perf_counter()
        reply = client.messages.create(
            model=MODEL,
            max_tokens=MAX_TOKENS,
            tools=TOOLS,
            messages=conversation,
        )
        secs = round(time.perf_counter() - started, 2)
        tokens_in = getattr(reply.usage, "input_tokens", 0)
        tokens_out = getattr(reply.usage, "output_tokens", 0)
        row = {"turn": turn, "stop": str(reply.stop_reason), "branch": "", "tools": [], "in": tokens_in, "out":
            tokens_out, "secs": secs}
        log.append(row)
        say("came back      : " + describe_blocks(reply) + "  ·  " + str(tokens_in) + " in / " + str(tokens_out)
             + " out  ·  " + str(secs) + " s")
        say("stop_reason    : " + str(reply.stop_reason) + "  ->  " + STOP_REASONS.get(str(reply.stop_reason),
```

```
            STOP_REASONS["<other>"])[0])

        # Collect any text in this reply.
        text_this_turn = text_of(reply)
        if text_this_turn.strip():
            say("model text     :", text_this_turn)

        # ---- Every stop_reason value gets its own branch. No "else" that swallows. ----

        if reply.stop_reason == "tool_use":
            # The loop's fuel. Run every tool the model asked for, send all the results back.
            row["branch"] = "run tools, loop"
            conversation.append({"role": "assistant", "content": reply.content})
            results = []
            for block in reply.content:
                if block.type == "tool_use":
                    output = run_tool(block)
                    row["tools"].append(block.name)
                    say("TOOL CALL      : " + block.name + "(" + json.dumps(block.input) + ")   id=" +
                        str(block.id))
                    say("TOOL RESULT    : " + json.dumps(output) + "   (plain Python ran this; the model only
                        sees the JSON)")
                    results.append({
                        "type": "tool_result",
                        "tool_use_id": block.id,
                        "content": json.dumps(output),
                    })
            conversation.append({"role": "user", "content": results})
            say("case file      : now " + str(len(conversation)) + " messages (added the assistant's request and
                 "
                + str(len(results)) + " tool_result block(s) as one user turn)")
            continue
        if reply.stop_reason == "end_turn":
            # The natural exit. Return everything, including text saved from an earlier max_tokens cut.
            row["branch"] = "return text (exit)"
            say("branch         : end_turn is the natural exit. Returning " + str(len((text_so_far +
                text_this_turn).split())) + " words.")
            print_summary(log, "end_turn on turn " + str(turn), text_so_far + text_this_turn)
            return text_so_far + text_this_turn

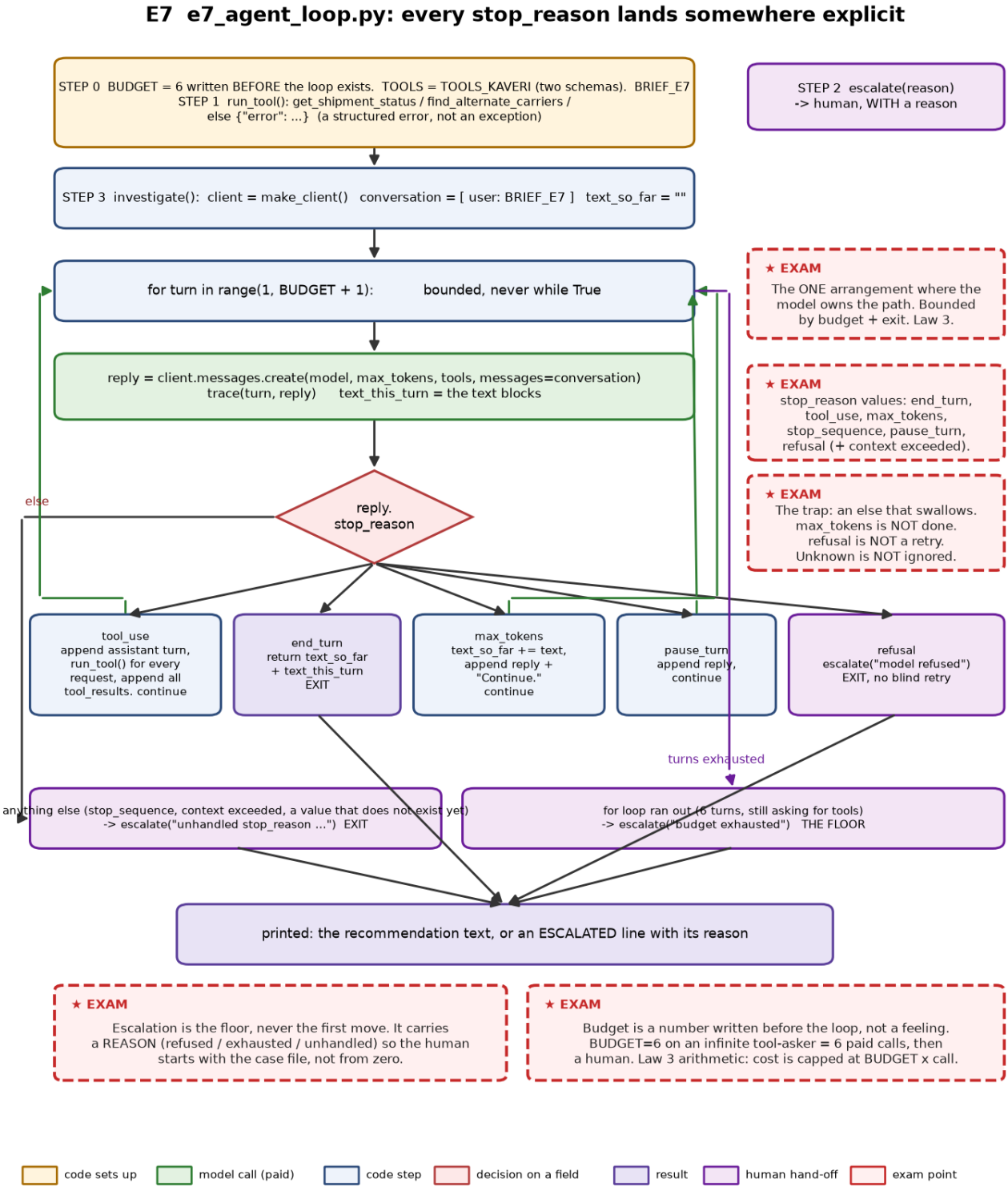
        if reply.stop_reason == "max_tokens":
            # Truncated is not done. Keep the partial text, ask the model to continue.
            row["branch"] = "keep partial, 'Continue.'"
            text_so_far = text_so_far + text_this_turn
            conversation.append({"role": "assistant", "content": reply.content})
            conversation.append({"role": "user", "content": "Continue."})
            say("branch         : cut off at max_tokens. Kept " + str(len(text_so_far.split())) + " words so
                far, asked the model to continue.")
            continue
        if reply.stop_reason == "pause_turn":
            # A server-side tool paused the turn. Send the reply back as it is to resume.
            row["branch"] = "resend to resume"
            conversation.append({"role": "assistant", "content": reply.content})
            say("branch         : server-side pause. Resending the reply unchanged so the server resumes.")
            continue

        if reply.stop_reason == "refusal":
            # The model declined. A different path, never a blind retry.
            row["branch"] = "ESCALATE (refused)"
            details = getattr(reply, "stop_details", None)
            why = (" category=" + str(getattr(details, "category", None))) if details else ""
            say("branch         : refusal" + why + ". No retry. The case goes to a human with the reason.")
            final = escalate("model refused" + why)
            print_summary(log, "refusal on turn " + str(turn), final)
            return final
        # Anything else: stop_sequence, context window exceeded, a value that does not exist yet.
        row["branch"] = "ESCALATE (unhandled)"
        say("branch         : a stop_reason this loop has no branch for. Escalating rather than guessing.")
        final = escalate("unhandled stop_reason " + str(reply.stop_reason))
        print_summary(log, "unhandled stop_reason on turn " + str(turn), final)
        return final

    # The for loop ran out. The case is still open. A human takes it, with the reason.
    print()
    say("BUDGET         : " + str(BUDGET) + " turns spent and the model is still asking. The floor catches it.")
    final = escalate("budget exhausted")
    print_summary(log, "budget exhausted after " + str(BUDGET) + " turns", final)
    return final
if __name__ == "__main__":
```

```
print("# E7 · agent loop · " + ("DRY_RUN" if DRY_RUN else "LIVE · " + MODEL) + " · scenario=" + SCENARIO + "
     · BUDGET=" + str(BUDGET))
print_legend()
investigate()
```

# E7 The agent loop: the flowchart




### Exam points on this chart

★ The one arrangement where the model owns the path, bounded by budget and exit. Law 3.
★ Values: end_turn, tool_use, max_tokens, stop_sequence, pause_turn, refusal, and context exceeded.
★ The trap is an else that swallows. max_tokens is not done, refusal is not a retry, escalation carries a reason.

# E8 Re-planning: the code

E5 plus a check after every finding: does the remaining plan still hold?

```
# E8 · Re-planning: the bridge is out, a finding can change the plan  (objective 1.5, dynamic decomposition)
#
#   Same three moves as E5 (make_plan -> run_worker -> synthesise), plus after EVERY finding the foreman is
    asked:
#   "does the remaining plan still make sense?" If not, drop some tasks and add others.
#   A second budget, MAX_NEW, caps how many tasks the plan may grow by. Law 3 applies to plans, not just turns.
#
#   The output is laid out so the plan can be WATCHED changing: the plan as a table at the start, one block per
#   run (the task, what it needed, the finding, the foreman's verdict), the remaining plan reprinted after every
#   replan with a growth meter, and a ledger at the end of what was planned, run, dropped and added.
#
# Run on battery:  DRY_RUN=1 python e8_replanning.py       Run live:  python e8_replanning.py
import json
import textwrap
from common import ask_json, MODEL, BRIEF, DRY_RUN
from e5_orchestrator_workers import make_plan, run_worker, synthesise     # E5's three moves, reused
from e5_orchestrator_workers import heading, block, show_plan              # E5's printing helpers, reused


# ------------------------------------------------------------------
# STEP 0. The growth budget
# ------------------------------------------------------------------

MAX_NEW = 3                              # at most 3 tasks may ever be added to the plan
# ------------------------------------------------------------------
# STEP 1. The new move: after a finding, ask whether the remaining plan still holds.
#         The foreman sees the new finding and the REMAINING tasks, not the whole history.
# ------------------------------------------------------------------

def check_plan(finding, remaining_tasks):
    question = (
        "A worker just reported this finding:\n" + finding + "\n\n"
        "These subtasks are still to be done:\n" + json.dumps(remaining_tasks) + "\n\n"
        "Does the remaining plan still make sense given the finding? Reply with JSON only:\n"
        '  {"ok": true}\n'
        "or\n"
        '  {"ok": false, "drop": [ids of tasks that no longer make sense], "add": ["short name of a new task",
            ...]}'
    )
    return ask_json(MODEL, question)
# ------------------------------------------------------------------
# STEP 2. Printing helpers specific to re-planning. They only print.
# ------------------------------------------------------------------

def growth_meter(grew):
    """grew 2 of 3  [##.]"""
    return "grew " + str(grew) + " of " + str(MAX_NEW) + "  [" + "#" * grew + "." * (MAX_NEW - grew) + "]"


def show_remaining(tasks, grew):
    if tasks:
        print("    remaining plan (" + str(len(tasks)) + " task" + ("" if len(tasks) == 1 else "s") + ") · " +
            growth_meter(grew))
        show_plan(tasks)
    else:
        print("    remaining plan: empty · " + growth_meter(grew))
def ledger(initial, order, dropped, added, grew, checks):
    heading("LEDGER · what was planned, what actually ran")
    print("  initial plan      : " + ", ".join(str(t["id"]) for t in initial) + "   (" + str(len(initial)) + "
        tasks from the foreman)")
    print("  ran, in order     : " + ", ".join(str(i) for i in order) + "   (" + str(len(order)) + " worker
        calls)")
    print("  dropped unrun     : " + (", ".join(str(i) for i in dropped) if dropped else "none"))
    print("  added on the way  : " + (", ".join(str(t["id"]) + " " + repr(t["task"]) for t in added) if added
        else "none"))
    print("  growth budget     : " + growth_meter(grew) + ("  EXHAUSTED" if grew >= MAX_NEW else "  not
        exhausted"))
    print("  plan checks       : " + str(checks["asked"]) + " asked · " + str(checks["ok"]) + " said OK · "
          + str(checks["replan"]) + " replanned · " + str(checks["blocked"]) + " blocked by the budget")
    print("  calls made        : 1 plan + " + str(len(order)) + " workers + " + str(checks["asked"]) + " checks
        + 1 synthesis = "
          + str(2 + len(order) + checks["asked"]))
# ------------------------------------------------------------------
# STEP 3. The loop with re-planning. Identical in logic to the plain version; the prints are the only addition.
```

```
# ------------------------------------------------------------

def replan(brief):
    heading("FOREMAN · the initial plan (1 call)")
    tasks = make_plan(brief)                              # E5's opening move
    initial = list(tasks)
    show_plan(tasks)

    findings = {}
    grew = 0                                              # how many tasks have been added so far
    next_id = max(task["id"] for task in tasks) + 1       # fresh ids for added tasks
    order, dropped, added = [], [], []                    # for the ledger only
    checks = {"asked": 0, "ok": 0, "replan": 0, "blocked": 0}
    heading("WORK · one worker per task, then 'does the rest of the plan still hold?' after EVERY finding")
    run_number = 0
    while tasks:                                          # keep going until the task list is empty
        current = tasks.pop(0)                            # take the first remaining task
        run_number = run_number + 1
        result = run_worker(current, findings)            # E5's worker call
        findings[current["id"]] = result
        order.append(current["id"])
        print()
        block("Run " + str(run_number) + " · task " + str(current["id"]) + " · needs "
              + (str(current["needs"]) if current["needs"] else "nothing") + " · " +
                  textwrap.shorten(current["task"], 70), result)
        verdict = check_plan(result, tasks)               # the walk-and-look
        checks["asked"] = checks["asked"] + 1

        if verdict.get("ok", True):
            checks["ok"] = checks["ok"] + 1
            print("    foreman: plan still holds · " + str(len(tasks)) + " task" + ("" if len(tasks) == 1 else
                "s") + " left · " + growth_meter(grew))
            continue                                      # plan still holds, move on

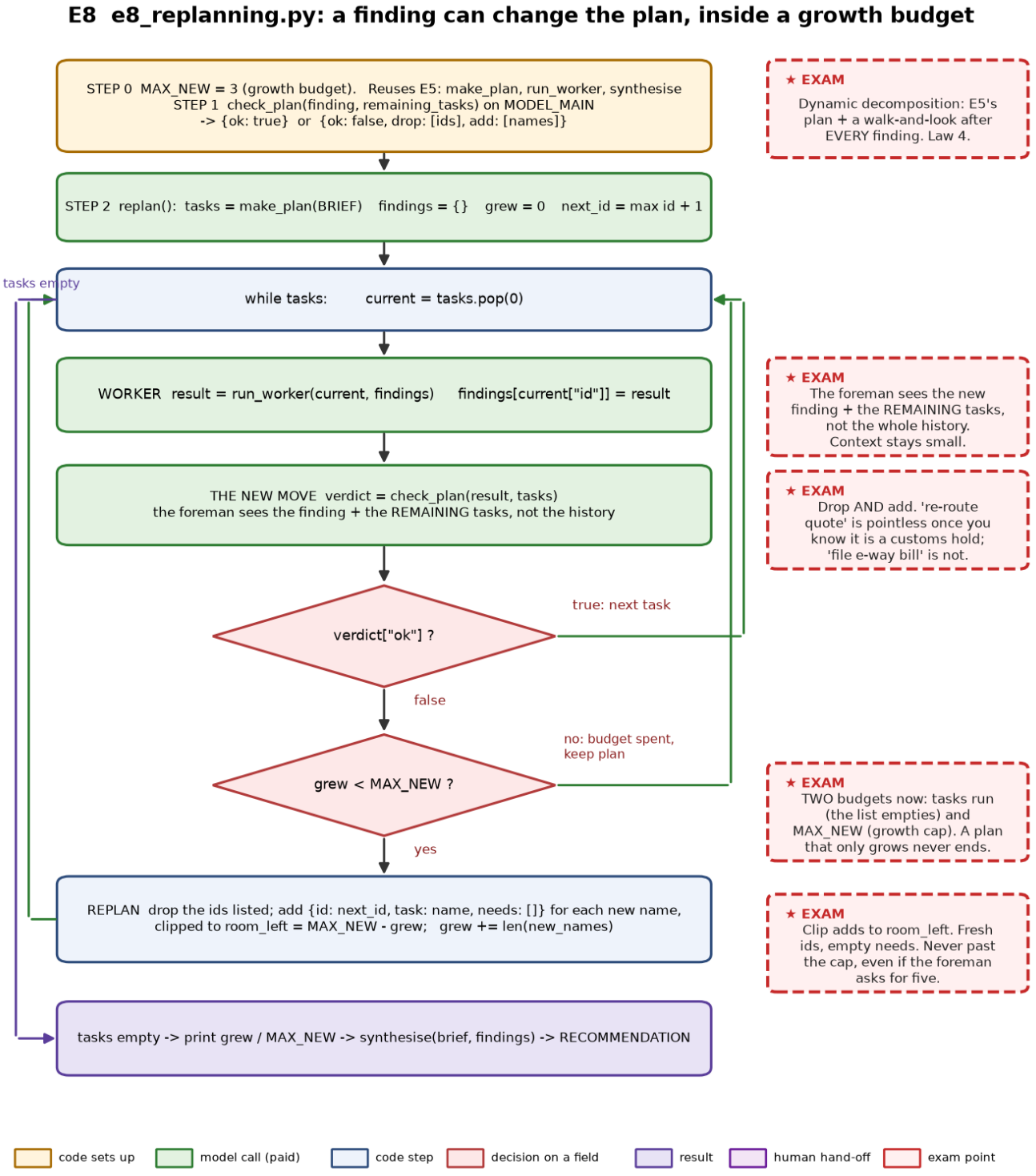
        if grew >= MAX_NEW:
            checks["blocked"] = checks["blocked"] + 1
            print("    foreman: wants to replan (drop " + str(verdict.get("drop", [])) + ", add " +
                str(verdict.get("add", []))
                  + ") · BLOCKED: the growth budget is spent · " + growth_meter(grew))
            continue
        # Drop the tasks the foreman says no longer make sense.
        drop_ids = set(verdict.get("drop", []))
        dropped.extend(task["id"] for task in tasks if task["id"] in drop_ids)
        tasks = [task for task in tasks if task["id"] not in drop_ids]

        # Add the new ones as proper task records, but never past the cap.
        room_left = MAX_NEW - grew
        new_names = verdict.get("add", [])[:room_left]
        cut = len(verdict.get("add", [])) - len(new_names)
        for name in new_names:
            tasks.append({"id": next_id, "task": name, "needs": []})
            added.append(tasks[-1])
            next_id = next_id + 1
        grew = grew + len(new_names)
        checks["replan"] = checks["replan"] + 1
        print("    foreman: REPLAN · dropped " + str(sorted(drop_ids)) + " · added " + str(new_names)
              + ("  (" + str(cut) + " more suggested, cut by the budget)" if cut else ""))
        show_remaining(tasks, grew)
    ledger(initial, order, dropped, added, grew, checks)

    heading("SYNTHESIS · sees every finding, writes one recommendation (1 call)")
    recommendation = synthesise(brief, findings)          # E5's closing move
    block("", recommendation, indent="  ")
    return recommendation


if __name__ == "__main__":
    print("# E8 · re-planning · " + ("DRY_RUN" if DRY_RUN else "LIVE · " + MODEL) + " · MAX_NEW=" +
        str(MAX_NEW))
    replan(BRIEF)
```

# E8 Re-planning: the flowchart




## Exam points on this chart

★ Dynamic decomposition. Law 4. The foreman sees the new finding and the remaining tasks only.
★ Drop and add, with fresh ids and empty needs.
★ Two budgets: the task list empties, and MAX_NEW caps growth. A plan that only grows never ends.

# Appendix standin.py: the dry-run stand-in

This is what answers when DRY_RUN=1. It reads the conversation and replies in character so the exhibits can run their real control flow without a key. It is not part of the patterns being taught; it is the battery.

```
# standin.py  -  a scripted stand-in for the Messages API, used when DRY_RUN=1.
#
# You do not need to read this file to learn the patterns. It exists so every exhibit can run its real
# control flow (the same stop_reason branches, the same replayed transcript) with zero paid calls.
# It reads the conversation it is given and answers in character:
#   - with tools attached, it asks for the first tool that has no result yet, then gives a finale
#   - without tools, it matches a few words in the prompt and returns a canned reply
# D1_SCENARIO=default|truncate|refuse|exhaust changes the finale so E7's branches can be watched.

import itertools
import json
import os
import re
from types import SimpleNamespace
SCENARIO = os.environ.get("D1_SCENARIO", "default")

# ---------------------------------------------------------------
# The canned world: fixed texts the stand-in hands back
# ---------------------------------------------------------------

DRAFT_FIRST = ("Dear Ms Deshmukh, thank you for reaching out to us regarding consignment KF-2481, and please
    accept "
               "our sincere apologies for the delay and for the lack of detail on the portal. I have checked the
                    "
               "consignment personally and can confirm it is currently held at the Chennai inland container
                   depot "
               "for a customs document check; specifically, the e-way bill attached to the shipment did not
                   match "
               "the invoice value. Our team is re-filing the corrected bill today and we expect the depot to
                   release "
               "the cartons within twenty-four hours of acceptance. Because of the inconvenience this has caused
                    you "
               "and your buyer, we will refund your shipping charges in full and upgrade your next three
                   shipments "
               "to express at no cost. I will send you the revised ETA personally as soon as the officer
                   confirms "
               "release. Warm regards, KAVI for Kaveri Freight.")
CRITICISM = ("Over 120 words, and 'we will refund your shipping charges in full' promises compensation the desk
    "
             "has not approved.")
DRAFT_GOOD = ("Dear Ms Deshmukh, thank you for your message about KF-2481, and I am sorry for the delay and the
    "
              "silence on the portal. The consignment is held at the Chennai depot for a customs document check:
                   "
              "the e-way bill did not match the invoice value. We are re-filing the corrected bill today, and
                  the "
              "depot normally releases within a day of accepting it. I will send you the revised delivery time
                  the "
              "moment the officer confirms. Your buyer may quote reference KF-2481 to us directly if that helps.
                   "
              "Regards, KAVI for Kaveri Freight.")
PLAN_JSON = ('[{"id": 1, "task": "status check", "needs": []}, '
             '{"id": 2, "task": "carrier options", "needs": [1]}, '
             '{"id": 3, "task": "re-route quote", "needs": [2]}]')
FINDINGS = {
    "status check":    "CUSTOMS_HOLD at Chennai depot; missing e-way bill; 3 days late",
    "carrier options": "BlueDart 2 days, Delhivery 3 days on Pune-Chennai; both need customs release first",
    "re-route quote":  "Re-route quote Rs 2,400 via BlueDart; only useful after customs release",
    "file e-way bill": "Corrected e-way bill filed per SOP 214 section 4; expected release within 24 h",
    "notify padma":    "Draft ready: revised ETA Saturday, goodwill note, no compensation promised",
}
RECOMMENDATION = ("RECOMMENDATION: file the e-way bill today; hold the re-route (costlier than the 24h release);
     "
                  "send Padma the revised ETA with a goodwill note.")
FINAL_E7 = ("KF-2481 is held at customs for a missing e-way bill. Options: file the bill (fastest, release ~24
    h), "
            "or re-route via BlueDart/Delhivery once released. Recommend filing today and telling Padma the
                revised ETA.")
_votes = itertools.count()     # E4 voting: APPROVE, REJECT, APPROVE -> majority APPROVE
_rubric = itertools.count()    # E6: the first evaluation criticises, the second says PASS
def canned_text(system, prompt):
    """Match a few words in the system prompt and the user prompt, return the matching canned reply."""
```

```
    p = (system or "").lower() + "\n" + prompt.lower()

    if "one word only" in p:                          # E2 desk 1, E3 nurse
        return "customs"
    if "approve or reject" in p:                      # E4 voting
        return "REJECT" if next(_votes) % 3 == 1 else "APPROVE"
    if "eta for" in p:                                # E4 scouts
        m = re.search(r"via (\w+)", prompt)
        carrier = m.group(1) if m else "carrier"
        return carrier + ": Thu 16:00, held at customs until release"
    if "rubric:" in p:                                # E6 editor
        return CRITICISM if next(_rubric) == 0 else "PASS"
    if "revise the draft" in p:                       # E6 writer, second pass
        return DRAFT_GOOD
    if "break this into" in p:                        # E5 foreman
        return PLAN_JSON
    if "remaining plan" in p:                         # E8 check after a finding
        if "customs_hold" in p and "re-route" in p:
            return json.dumps({"ok": False, "drop": [3], "add": ["file e-way bill", "notify Padma of revised
                ETA"]})
        return json.dumps({"ok": True})
    if "recommendation for meera" in p:               # E5 and E8 synthesis
        return RECOMMENDATION
    if "subtask:" in p:                               # E5 and E8 workers
        m = re.search(r"subtask: (.+)", p)
        name = m.group(1).strip() if m else ""
        for key, value in FINDINGS.items():
            if name.startswith(key):
                return value
        return "no finding"
    if "draft a courteous reply" in p:                # E2 desk 2
        return DRAFT_GOOD
    if "exception email" in p:                        # E6 first draft (deliberately too long, promises a
        refund)
        return DRAFT_FIRST
    if "refund disputes" in p:                        # E3 wards
        return "Refund ward: claim acknowledged; verifying the damage note and insurance cover before any
            decision. No promise made."
    if "customs holds" in p:
        return "Customs ward: KF-2481 is held for an e-way bill mismatch. Next steps: re-file the corrected bill
             (SOP 214 section 4), expect release within 24 h."
    if "delay apologies" in p:
        return "Delay ward: We are sorry KF-2481 is late. New ETA: Saturday. We will confirm once the depot
            releases it."
    return "dry-run reply"
# ---------------------------------------------------------------
# Reply objects shaped like the SDK's: .content (a list of blocks), .stop_reason, .usage
# ---------------------------------------------------------------

def _block(**fields):
    return SimpleNamespace(**fields)


def _reply(blocks, stop_reason, **extra):
    return SimpleNamespace(content=blocks, stop_reason=stop_reason,
                           usage=SimpleNamespace(input_tokens=0, output_tokens=0), **extra)
def _field(block, name):
    """Read a content-block field whether the block is an SDK object or a plain dict. The API takes both."""
    if isinstance(block, dict):
        return block.get(name)
    return getattr(block, name, None)


class _FakeMessages:
    def __init__(self):
        self.calls = 0

    def create(self, *, model, max_tokens, messages, system=None, tools=None, **_):
        self.calls = self.calls + 1
        if tools:
            return self._with_tools(messages, system or "", tools)
        # No tools: answer the last user message from the canned world.
        last = messages[-1]["content"]
        if isinstance(last, str):
            prompt = last
        else:
            prompt = " ".join(b.get("text", "") for b in last if isinstance(b, dict))
        return _reply([_block(type="text", text=canned_text(system, prompt))], "end_turn")
    def _with_tools(self, messages, system, tools):
```

```
        # Read the stateless transcript: which tool requests already have a result?
        id_to_name = {}
        answered = set()
        for message in messages:
            content = message["content"]
            if message["role"] == "assistant" and not isinstance(content, str):
                for block in content:
                    if _field(block, "type") == "tool_use":
                        id_to_name[_field(block, "id")] = _field(block, "name")
            if message["role"] == "user" and isinstance(content, list):
                for block in content:
                    if isinstance(block, dict) and block.get("type") == "tool_result":
                        answered.add(id_to_name.get(block["tool_use_id"]))
        transcript = json.dumps(messages, default=str)
        found = re.search(r"KF-\d{4}", transcript)
        consignment_id = found.group() if found else "KF-2481"

        if SCENARIO == "exhaust":                     # a detective who never stops asking for the phone
            return _reply([_block(type="tool_use", id="toolu_dry_" + str(self.calls),
                                  name="get_shipment_status", input={"consignment_id": consignment_id})],
                              "tool_use")

        for tool in tools:                            # the first tool with no result yet gets called
            if tool["name"] not in answered:
                if tool["name"] == "get_shipment_status":
                    tool_input = {"consignment_id": consignment_id}
                else:
                    tool_input = {"lane": "Pune-Chennai"}
                return _reply([_block(type="tool_use", id="toolu_dry_" + str(self.calls),
                                      name=tool["name"], input=tool_input)], "tool_use")
        # Every tool answered: the finale, steered by the scenario.
        if SCENARIO == "refuse":
            return _reply([], "refusal",
                          stop_details=SimpleNamespace(type="refusal", category="general_harms",
                                                       explanation="scripted dry-run refusal"))

        if "triage" in system.lower():
            final = '{"category": "customs", "urgency": "high"}'
        else:
            final = FINAL_E7

        continued = False
        for message in messages:
            if message["role"] == "user" and message["content"] == "Continue.":
                continued = True
        if SCENARIO == "truncate" and not continued:
            cut = len(final) // 2
            return _reply([_block(type="text", text=final[:cut])], "max_tokens")
        if SCENARIO == "truncate" and continued:
            return _reply([_block(type="text", text=final[len(final) // 2:])], "end_turn")
        return _reply([_block(type="text", text=final)], "end_turn")


class FakeClient:
    def __init__(self):
        self.messages = _FakeMessages()
```